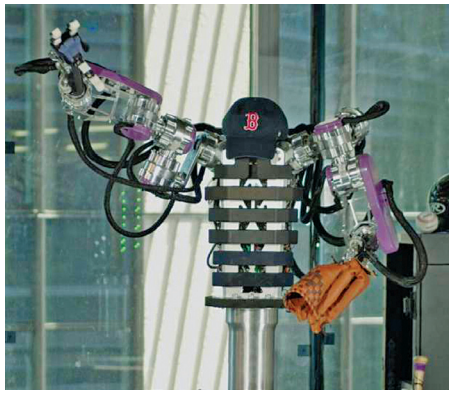
Wireless mesh targets large-scale IoT networks

Neomesh is a decentralised wireless mesh technology designed to connect large numbers of low-power IoT devices while reducing deployment complexity. Nodes communicate bidirectionally and automatically form time-synchronised communication paths, with gateways linking sensor data to cloud services. The technology also supports LoRa modulation for long-range, low-power connectivity and can be integrated through dedicated modules or licensed for different chipsets. Demonstrated by NeoCortec at The Things Conference 2026, it targets industrial and building IoT, where battery-powered sensors benefit from scalable connectivity, automatic network formation, and reduced maintenance requirements.

Dual-arm robot mimics human movements

Researchers at the Robotics and AI Institute in Cambridge, Massachusetts, have developed AthenaZero, a dual-arm robot that can throw, catch, and hit balls with human-like movements. Its low-inertia mechanical design, custom actuators, and backdrivable joints allow the arms to respond to external forces and adjust stiffness during physical interactions. Tested on baseball-inspired tasks, the system demonstrates coordinated movement, rapid response, and controlled contact. The technology could support industrial assembly, object handling and safer human-robot interaction, advancing robotic manipulation beyond conventional rigid, position-focused designs.



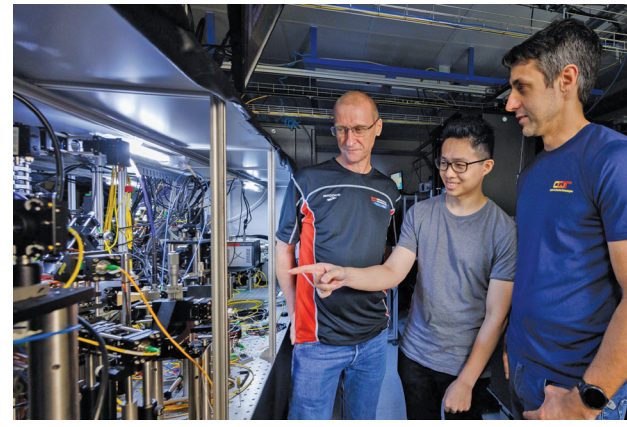
AthenaZero throwing a baseball (Credit: Morgan et al)

New catalyst uses sunlight to extract hydrogen from water

A porous metal-organic framework developed by researchers at Oregon State University can use light to drive hydrogen production from water, reaching 938µmol of hydrogen per hour per gram of catalyst. Called BVR-19-Zn, the material uses a sulphur-containing organic linker rather than mainly its metal centre to handle light-driven charge transfer. This design could reduce dependence on expensive co-catalysts in solar-fuel production. The team synthesised the material in water at room temperature and demonstrated repeated hydrogen generation over six two-hour cycles. The approach could support future solar-hydrogen systems, although practical water splitting remains to be demonstrated.

The world's most accurate atomic clock

Researchers at the National University of Singapore (NUS) have developed the world's most accurate optical atomic clock, achieving a reported uncertainty of 1×10^{-19}. Built at NUS' Centre for Quantum Technologies, the clock uses singly charged lutetium-176 ions and a laser locked to an atomic transition. Lutetium's low sensitivity to temperature and magnetic fields reduces sources of measurement error. Two independently built clocks agreed to an uncertainty of 5.7×10^{-19} after 200 hours of comparison. Led by Associate Professor Murray Barrett, the work could improve satellite navigation, geodesy and fundamental-physics measurements while supporting efforts to redefine the second using optical standards.



From left: Associate Prof. Murray Barrett, PhD student Michael Lee and Senior Research Scientist Kyle Arnold with the lutetium-ion optical clock at NUS (Image: NUS)

Silicon-carbon batteries boost phone capacity

Silicon-carbon anodes are helping smartphone batteries cross 10,000mAh without a proportional increase in device thickness. The technology adds silicon to conventional graphite, raising theoretical anode capacity from about 372mAh/g for graphite to roughly 4200mAh/g for silicon. A carbon structure accommodates silicon's expansion during charging, improving electrode stability. Realme's P4 Power 5G, for example, uses a third-generation silicon-carbon anode in its 10,001mAh battery. Advances in cell packaging add further capacity. However, higher costs, charging-related ageing and limited long-term field data remain challenges.